Progress on SU(3) downsampling and upsampling

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I. REFERENCE CONFIGURATIONS AND MEASUREMENTS

We study factor-two reduction from a 24^4 fine lattice to 12^4, followed by stochastic reconstruction of a fine field. The full pipeline is

$$U \xrightarrow{S_{\boldsymbol{\theta}}} V \xrightarrow{B} C, \qquad (C, z) \xrightarrow{\mathcal{L}} V_0 \xrightarrow{F_1} \widehat{V} \xrightarrow{F_2} \widehat{U}. \tag{1.1}$$

Here U denotes the complete reference fine-link configuration, with each link in $SU(3)$. $S_{\boldsymbol{\theta}}$ is our stout-type smoothing plus $SU(3)$ projection, and $\boldsymbol{\theta}$ contains its eleven fitted coefficients. Specifically, it mixes covariant-Laplacian polynomial, hook and local plaquette terms, then projects the resulting matrices by polar decomposition; it is not the standard exponential stout update. $V = S_{\boldsymbol{\theta}}(U)$ is the smoothed configuration on the same fine lattice. B is two-link blocking: $C_\mu(X) = V_\mu(2X)V_\mu(2X + \hat{\mu})$, where X is a coarse site and $\hat{\mu}$ a unit fine-lattice step. $C = B(V)$ is the coarse field.

$\mathcal{L}$ denotes the stochastic *lift* from coarse to fine links, before either learned flow. Its input z consists of 60 independent Haar-distributed $SU(3)$ detail matrices per coarse cell. Four of these, A_μ, set the first retained links; the second links are $A_\mu^\dagger C_\mu$, and the other 56 fine links receive the remaining details. Thus $V_0 = \mathcal{L}(C, z)$ is a random fine field satisfying $B(V_0) = C$; the noise is group-valued, not added to matrix entries.

F_1 and F_2 are learned invertible, gauge-equivariant field transforms. F_1 produces the smeared proposal $\widehat{V}$ while preserving its retained coarse products; F_2 produces the final fine proposal $\widehat{U}$. Hats distinguish generated fields from reference fields. Generation requires only C, z; the paired fine and smoothed configurations supply training and evaluation targets. The next sections specify the smoother and the two transforms.

A. Reference ensembles

The fine and coarse references each contain 300 configurations, generated independently with periodic boundaries and Wilson action

$$S_W[U] = \beta \sum_{x,\mu<\nu} \left(1 - \frac{1}{3} \operatorname{Re} \operatorname{Tr} U_{\mu\nu}(x)\right). \tag{1.2}$$

The fine ensemble has $L = 24$, $\beta = 6.20$; the coarse ensemble has $L = 12$, $\beta = 5.80$. Each chain starts hot, receives 1000 warmup steps, and saves a configuration every 100 steps. A step is one heatbath hit plus four overrelaxation hits, followed by reunitarization. Seed 1234 supplies the first 50 configurations and seed 1235 the next 250, with a separate hot start and warmup for each stream. QUDA uses single precision; full 3×3 links are saved in NERSC IEEE32 little-endian format without gauge fixing.

The static-potential check uses 20 spatial APE steps ($\alpha = 0.5$), Wilson loops with $r/a, t/a = 1, \ldots, 6$, an exponential fit over $t/a = 2, \ldots, 5$, and a Cornell fit over $r/a = 2, \ldots, 6$. With 1000 bootstrap samples and $r_0 = 0.5$ fm, it gives $a_f = 0.0722350(2175)$ fm and $a_c = 0.137312(1608)$ fm (Fig. 1). Their ratio $a_c/a_f \simeq 1.90$ is a measured scale ratio, distinct from the geometric blocking factor 2. Saved plaquette and APE-$W(2, 2)$ series have lag-one correlations between -0.055 and 0.070 and estimated statistical inefficiency 1. These checks concern the saved series, whose spacing is 100 update steps.

B. Observables and data split

The four observables are lattice averages of the normalized real traces of the plaquette, 1×2 rectangle and 2×2 square, plus the Polyakov observable. The first and third average over six unordered planes; the rectangle averages over twelve ordered direction pairs. For the Polyakov observable,

$$O_{\text{Poly}} = \frac{1}{4} \sum_{\mu=0}^{3} |\overline{P}_\mu|^2, \qquad \overline{P}_\mu = \frac{1}{|\Lambda|/L_\mu} \sum_{x_\perp} \frac{1}{3} \operatorname{Tr} \prod_{k=0}^{L_\mu - 1} U_\mu(x_\perp + k\hat{\mu}). \tag{1.3}$$

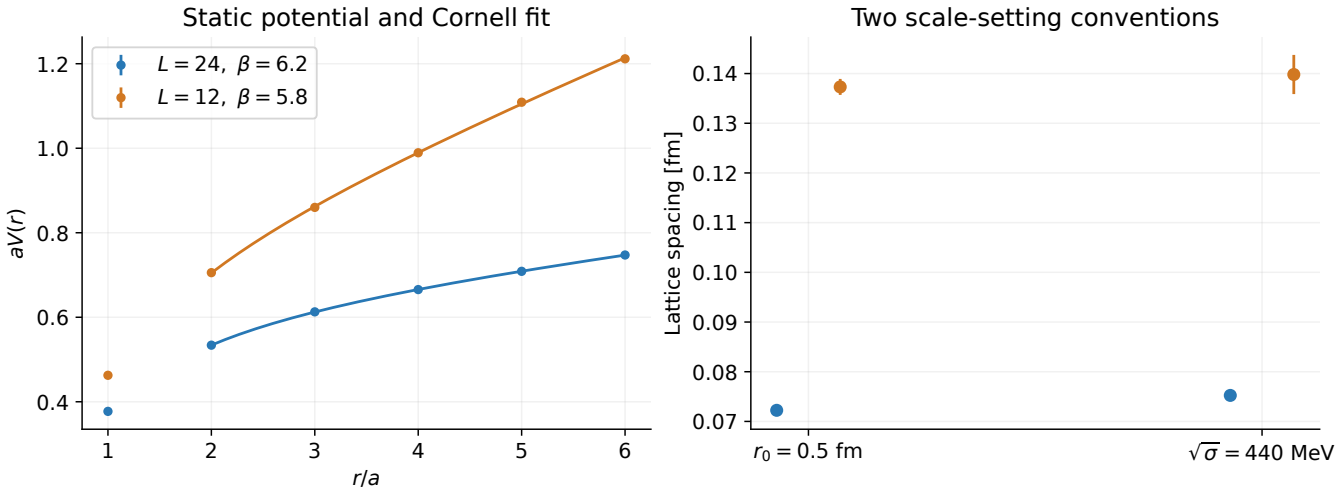


FIG. 1. Reference static potentials and lattice spacings from the Sommer scale and string tension. The measured spacing ratio is approximately 1.90.

The modulus is squared *after* spatial averaging of the complex trace, then averaged over all four winding directions.

Configuration indices $i = 0, \ldots, 299$ are split by $i \bmod 5$: $0, 1, 2$ for training, 3 for validation and 4 for testing, giving 180/60/60. Downsampling fits 60 fixed fine training configurations, chosen with seed 1234, against all 180 coarse training references. Upsampling uses paired $U_i, S(U_i)$ targets; its current subset is specified with the results below. Downsampling evaluation uses all 60 test configurations. No test target enters a fit.

For observable j, let $\bar{O}_j, s_j$ be the output sample mean and standard deviation, and $\bar{O}_{t,j}, s_{t,j}$ the corresponding target quantities. The subscript t means target. With n configurations,

$$\bar{O}_j = \frac{1}{n}\sum_i O_{i,j}, \qquad s_j^2 = \frac{1}{n-1}\sum_i (O_{i,j} - \bar{O}_j)^2. \tag{1.4}$$

The same definition applies to the target sample. These are distribution widths; the independent-sample mean error is $s_j/\sqrt{n}$. We report

$$\Delta_{\sigma,j} = \frac{\bar{O}_j - \bar{O}_{t,j}}{s_{t,j}}, \qquad R_{s,j} = \frac{s_j}{s_{t,j}}. \tag{1.5}$$

The stated distribution criterion is $|\Delta_{\sigma,j}| \le 1$ and $0.8 \le R_{s,j} \le 1.2$ for all four observables.

II. PROJECTED "STOUT" DOWNSAMPLING

A. Smoothing and blocking

Here "stout" denotes a projected smoothing family. Unlike standard exponential stout [3], it forms a raw transporter and projects it onto $SU(3)$. Early global path mixing was replaced by the current polynomial with hook and local-plaquette terms.

For a link-valued field $A_\mu(x)$, define the covariant transverse Laplacian

$$\begin{aligned}(L_U A)_\mu(x) = \sum_{\nu\neq\mu} \big[&U_\nu(x) A_\mu(x+\hat{\nu}) U_\nu^\dagger(x+\hat{\mu}) \\ &+ U_\nu^\dagger(x-\hat{\nu}) A_\mu(x-\hat{\nu}) U_\nu(x-\hat{\nu}+\hat{\mu}) - 2A_\mu(x)\big].\end{aligned} \tag{2.1}$$

All polynomial powers use the original transporters U; in particular, $L_U U$ is the six-staple sum minus $6U$. The fitted

TABLE I. Physical coefficients of the frozen smoother.

Order	a_n	b_{n-1}	c_n
1	0.1984211682	0.03923634322	0.03779125486
2	−0.02362395915	0.02501181641	−0.02158371669
3	−0.002876974930	0.002998340021	0.002722206164
4	$-6.659929755 \times 10^{-5}$	—	$-2.232800723 \times 10^{-4}$

TABLE II. Downsampling on 60 test fine and coarse configurations.

Observable	Δ_σ	R_s
Plaquette	0.208	1.259
W_{12}	0.288	1.033
W_{22}	0.294	0.970
Polyakov	−0.101	0.725

raw smoother is

$$W_\mu = U_\mu + \sum_{n=1}^{4}(a_n + f_\mu c_n)(L_U^n U)_\mu + \sum_{m=0}^{2} b_m (L_U^m \mathcal{H})_\mu, \tag{2.2}$$

$$f_\mu = \frac{1}{6}\,\mathrm{Re}\,\frac{1}{3}\,\mathrm{Tr}[U_\mu^\dagger (L_U U)_\mu]. \tag{2.3}$$

The gauge-invariant f_μ is the mean adjacent plaquette trace minus one. The hook field $\mathcal{H}$ is the unnormalized sum of twelve P5/P6 paths: $s\mu\mu(-s)(-\mu)$ and $(-\mu)s\mu\mu(-s)$ for $s = \pm\nu$, $\nu \neq \mu$. Thus there are eleven fitted coefficients: four a_n, three b_m and four c_n.

For the SVD $W = A\Sigma B^\dagger$, let $Q = AB^\dagger$. Smoothing and blocking are then

$$V_\mu = e^{-i \arg\det Q/3} Q, \qquad C_\mu(X) = V_\mu(2X) V_\mu(2X + \hat{\mu}). \tag{2.4}$$

Only even anchors are retained. The implementation adds $10^{-6}\,\mathrm{diag}(1,2,3)$ before SVD to regularize derivatives; this fixed-colour perturbation introduces a small covariance defect.

B. Fitting and results

The initial four-coefficient polynomial was chosen by matching weak-field Wilson propagator shapes after blocking. With left-endpoint convention $U_\mu = 1 + ie^{ik\cdot x} A_\mu(k) + O(A^2)$, its kernel is

$$\omega_{\mu\nu}(k) = \delta_{\mu\nu} + \sum_{n=1}^{4} a_n d_\mu(k)^{n-1} L_{\mu\nu}(k), \tag{2.5}$$

where $d_\mu = \sum_{\rho\neq\mu}(2\cos k_\rho - 2)$, $L_{\mu\mu} = d_\mu$ and $L_{\mu\nu} = (1-e^{ik_\mu})(1-e^{-ik_\nu})$ for $\mu \neq \nu$. The rowwise powers follow because $L_U U$ has no constant background. The propagator match sums the sixteen momenta $k = K/2 + \pi\ell$, $\ell \in \{0,1\}^4$, for each coarse momentum K. Hooks and local terms are added in the subsequent ensemble fit.

The final fit starts from the polynomial-plus-hook solution, with local coefficients zero. It minimizes four mean and four log-variance residuals:

$$r_{\mu,j} = \frac{\bar{O}_j - \bar{O}_{t,j}}{s_{t,j}\sqrt{1/60 + 1/180}}, \qquad r_{v,j} = \frac{\log[(s_j^2 + 10^{-12})/(s_{t,j}^2 + 10^{-12})]}{\sqrt{2/59 + 2/179}}. \tag{2.6}$$

Trust-region least squares uses analytic ensemble derivatives and scaled coordinates $12^n a_n$, $12^{m+1} b_m$, $12^n c_n$ to balance polynomial orders. Cached basis fields at retained bond positions accelerate the fit. The objective decreases from 304.015 to 5.6853, stopping at the budget of 40 evaluations. Table I gives the frozen parameters.

All four mean shifts pass Eq. (1.5), but the plaquette and Polyakov widths fail the 20% window (Table II, Fig. 2).

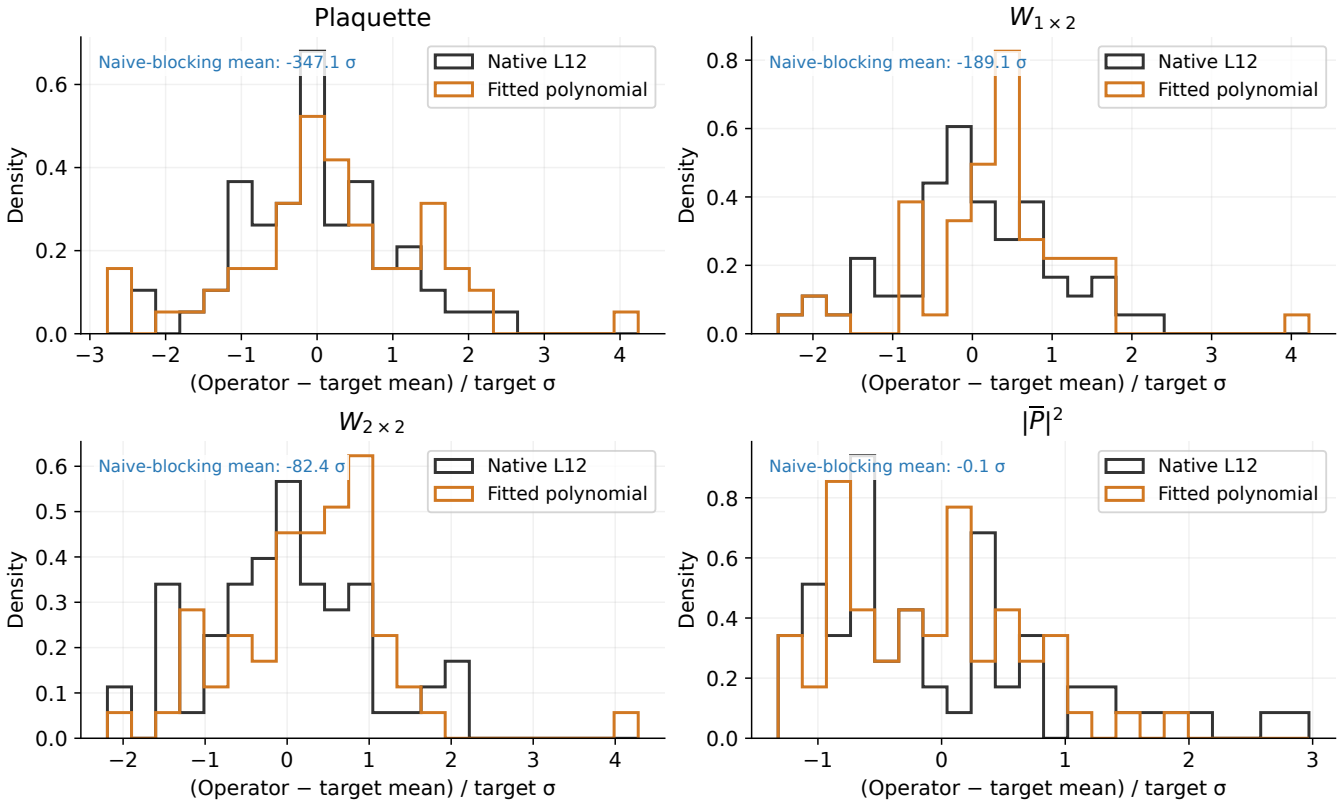


FIG. 2. Downsampled and native coarse distributions on 60 test configurations, standardized by each target mean and width.

III. UPSAMPLING WITH HAAR DETAILS AND TWO FLOWS

A. Coarse constraints and random details

Each coarse cell contains 2^4 fine sites, hence 64 fine links. Fixing four coarse two-link products leaves 60 independent Haar $SU(3)$ links (480 real detail dimensions). On retained bonds the initial lift sets

$$V_{0,\mu}(2X) = A_\mu(X), \qquad V_{0,\mu}(2X+\hat{\mu}) = A_\mu(X)^\dagger C_\mu(X), \tag{3.1}$$

with A_μ Haar, while the other 56 links are independently Haar. This gives $B(V_0) = C$. Noise is group-valued, rather than added to link matrices. The lift has unit Haar Jacobian and zero normalized-Haar base log-density.

Flow 1 freezes both links on retained bonds and transforms the other 56 links, preserving $B(\widehat{V}) = C$. Flow 2 transforms all fine links to learn the transport from smeared to fine fields. It does not impose $B(S(\widehat{U})) = C$. This two-stage conditional construction is motivated by cascade upscaling [4].

B. Local coupling and gauge equivariance

Each sweep updates one direction and one checkerboard parity at a time. The inactive context is $C^{\mathrm{ctx}} = C^{(1)} + \lambda C^{(3)}$, where $C^{(d)}$ averages six staples with transverse displacement d. Their lengths are three and seven links, respectively. Odd transverse displacement puts the same-direction link on the opposite parity, so no active link enters the context.

Following the invariant-conditioner and covariant-update principles of Refs. [1, 2], we use a local covariant-frame coupling rather than the exponential Wilson-loop-derivative update of Ref. [2]. Its inverse and Haar Jacobian follow from the subgroup maps below, not from that paper's coefficient bound. We construct an ordered colour frame

$$H = C^{\mathrm{ctx}}(C^{\mathrm{ctx}})^\dagger = F\Lambda F^\dagger, \qquad \overline{U} = F^\dagger U, \quad \overline{C} = F^\dagger C^{\mathrm{ctx}}. \tag{3.2}$$

Here Λ contains the ordered eigenvalues of H. In this frame, each colour pair (i,j) defines a quaternion from $M=\overline{UC}^{\dagger}$:

$$a=\tfrac{1}{2}(M_{ii}+M_{jj}^{*}),\qquad b=\tfrac{1}{2}(M_{ij}-M_{ji}^{*}),\qquad q=(a,b)/\sqrt{|a|^{2}+|b|^{2}}. \tag{3.3}$$

A $9\to 8\to 6$ network uses invariant staple norms and overlaps, the quaternion radius and site-parity bits to predict three scales and shapes. Although the radius depends on the active link, it is unchanged by its subgroup update; the other features depend only on inactive links or sites. Thus the conditioner is fixed along each subgroup fibre. Each normalized quaternion receives a radial and cubic deformation,

$$\tan(\theta'/2)=e^{\eta}\tan(\theta/2),\qquad u\mapsto g(u)=u+\alpha u(1-u^{2}). \tag{3.4}$$

Here $q_0=\cos\theta$ is the quaternion's real component and u its value after the radial step; the remaining components are rescaled to keep unit norm. The scale η is bounded by 2 and $|\alpha|<0.45$ makes g monotone. Both are suppressed by normalized eigenvalue gaps near degeneracy; zero or repeated-spectrum contexts use the identity update. The updated quaternion gives an $SU(2)$ rotation R, applied as $FRF^{\dagger}U$. The subgroup Haar Jacobians are analytic and add across couplings.

For endpoint gauge matrices G_x, G_y, links and contexts transform as $U\mapsto G_xUG_y^{\dagger}$, $C^{\mathrm{ctx}}\mapsto G_xC^{\mathrm{ctx}}G_y^{\dagger}$. The ordered frame transforms as $F\mapsto G_xFD$, where D contains arbitrary eigenvector phases. Consequently

$$M\mapsto D^{\dagger}MD,\qquad R\mapsto D^{\dagger}RD,\qquad FRF^{\dagger}U\mapsto G_x(FRF^{\dagger}U)G_y^{\dagger}. \tag{3.5}$$

The scalar conditioner and deformations commute with these phases, so they cancel from the physical update. Composing the masked updates preserves this equivariance. Haar details must be transformed with their fine-site endpoints when comparing samples under a gauge transformation.

The inverse reverses the coupling order, solves the cubic with eight Newton steps, and applies the radial map with $-\eta$. The current version keeps fields and conditioners in double precision. Whole-flow checks gate selected checkpoints, since deep contractions can amplify roundoff even when each local inverse is accurate. For the selected checkpoint, synchronized-noise 24^4 generation gives 9.23×10^{-13} maximum gauge error and 2.83×10^{-11} maximum roundtrip error. Full eight-dimensional local Haar Jacobians agree with the analytic values to 7.33×10^{-15}. The retained-product error is 1.12×10^{-14}; generated $SU(3)$ residuals are at most 5.79×10^{-6} with the cached single-precision constraints.

IV. TRAINING AND CURRENT UPSAMPLING RESULTS

The smoother is frozen and the flows have 8/2 sweeps. Adam updates all conditioner weights, biases and long-staple weights by backpropagation. The original fit uses 12 training and 8 validation configurations, with conditional likelihood, smeared moments, paired transport and fine moments trained for 4/20/3/20 epochs. Flow 1 uses $\log q_1(V\mid B(V))=\log J_{F_1^{-1}}(V)$ on its constrained Haar details; paired transport compares $F_2(S(U))$ with U. The Adam moment loss sums covariance-weighted squared mean residuals and squared log-variance residuals with width weight 1.

For the current calibration, Flow 1 is frozen at its stable uncalibrated checkpoint and Flow 2 starts from a previously calibrated checkpoint. This is not a direct continuation of the last plotted Adam checkpoint. Calibration trains the same Flow 2 in a restricted six-parameter subspace: for each sweep $s=1,2$, $\delta_{\eta,s}$ shifts all three raw scale output biases, $\delta_{h,s}$ shifts all three cubic-shape output biases, and $\delta_{\lambda,s}$ changes the long-staple weight. Each offset is shared across sites, directions and colour pairs; other network parameters stay fixed. The original coupling bounds and spectral gates remain in place, preserving the equivariant construction.

The inputs are cached actual Flow 1 proposals, not reference smeared fields. Fixed Haar draws use seed $1234+60000+i$ for configuration i, on 8 training and 8 validation configurations. Flow 2 changes the generated fine configurations themselves. Its calibration score is $\sum_j\Delta_{\sigma,j}^2+10\sum_j[\log((v_j+10^{-12})/(v_{t,j}+10^{-12}))]^2$, where $v=s^2$. This replaces the Adam mean metric by target-width normalization and increases the width weight to 10. Bounded least squares uses an eight-component residual and its finite-difference derivative with respect to six offsets (step 0.002), with bounds ±1.5 and at most 30 objective evaluations. These derivatives are not the Haar Jacobian; no lattice backpropagation or likelihood term is used. Finite-difference measurements are additional to the recorded objective count. Training residuals drive the search; selection minimizes validation score among whole-flow inverse passes with all validation mean shifts within 2 target widths. Six evaluations without validation improvement stop the recorded fit. The stricter test criterion in Eq. (1.5) is unchanged. Test configurations do not enter fitting or selection.

Adam could optimize these same offsets. The two stages differ in parameter subspace, objective weighting and starting checkpoint, so their histories do not establish an advantage of least squares over Adam. The small fixed-noise fitting sample can overfit; independent test configurations with fresh noise assess only the measured observables.

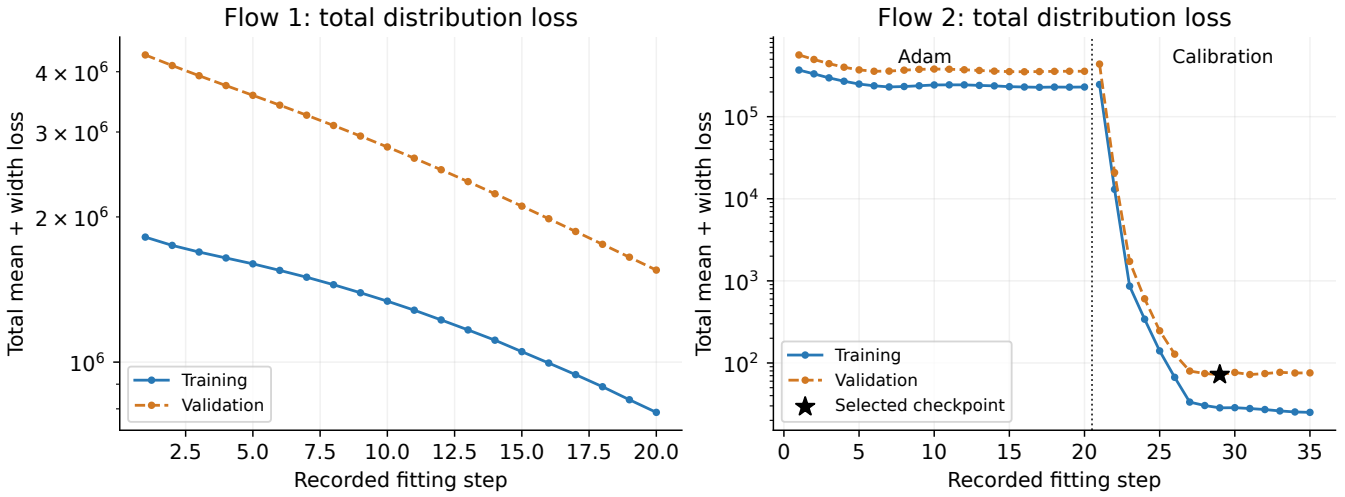


FIG. 3. Total mean-plus-width loss for each flow. Left: 20 epochs of Flow 1 smeared-distribution fitting. Right: 20 epochs of Flow 2 fine-distribution fitting, followed by 15 six-offset calibration evaluations with Flow 1 frozen. The dotted boundary separates the original covariance-weighted Adam objective from the calibration score defined above, with its different weighting and warm start; these segments are not joined. The star marks calibration evaluation 9. Adam uses a pool of 12 training and 8 validation configurations (6 training configurations per Flow 1 moment epoch); calibration uses 8 and 8.

TABLE III. Two-flow fine proposals on 60 test configurations.

Observable	Target mean	Generated mean	Δ_σ	R_s
Plaquette	0.613595	0.613799	1.068	1.727
W_{12}	0.410039	0.410177	0.513	1.245
W_{22}	0.21698	0.217128	0.425	0.914
Polyakov	8.6875×10^{-6}	8.12545×10^{-6}	−0.130	0.724

Figure 3 shows the recorded training and validation losses. Each flow has one panel showing the total distribution-fitting objective $L_{\text{total}} = L_{\text{mean}} + L_{\text{width}}$ over the four observables. Likelihood and paired-link warm-up losses are omitted because they have different units. No test loss enters these curves.

Evaluation uses one fresh draw for each of 60 test coarse configurations, obtained by downsampling fine reference configurations. No rethermalization, operator rescaling or test-target fitting is applied. Independent coarse heatbath inputs have not been evaluated.

For each test configuration U_i, the coarse input is $C_i = B(S_{\boldsymbol{\theta}}(U_i))$. The Flow 1 proposal $\widehat{V}_i = F_1(\mathcal{L}(C_i, z_i))$ is compared with the *smoothed fine-lattice reference* $V_i = S_{\boldsymbol{\theta}}(U_i)$, not the original fine field or the native coarse ensemble. The final proposal $\widehat{U}_i = F_2(\widehat{V}_i)$ is compared with the original U_i. Figure 4 separates these two comparisons, using each stage's own target mean and width. Flow 1 still fails to reproduce the smeared loop distributions. Flow 2 is fitted to its actual generated inputs and compensates enough to bring the four final observables roughly into overlap with their fine references. This does not establish the full fine-field distribution or identify Flow 2 with the inverse smoother.

The final fine output fails the stated distribution criterion (Table III, Fig. 4). Mean shifts outside one target width: Plaquette. Widths outside the 20% window: Plaquette, W_{12}, Polyakov. Intermediate smeared outputs fail the same criterion. Flow 2 with teacher inputs fail the same criterion. Flow 1 stays fixed. The fine-flow calibration used 15 objective evaluations, stopping on validation patience. Validation selects the numerically stable checkpoint.

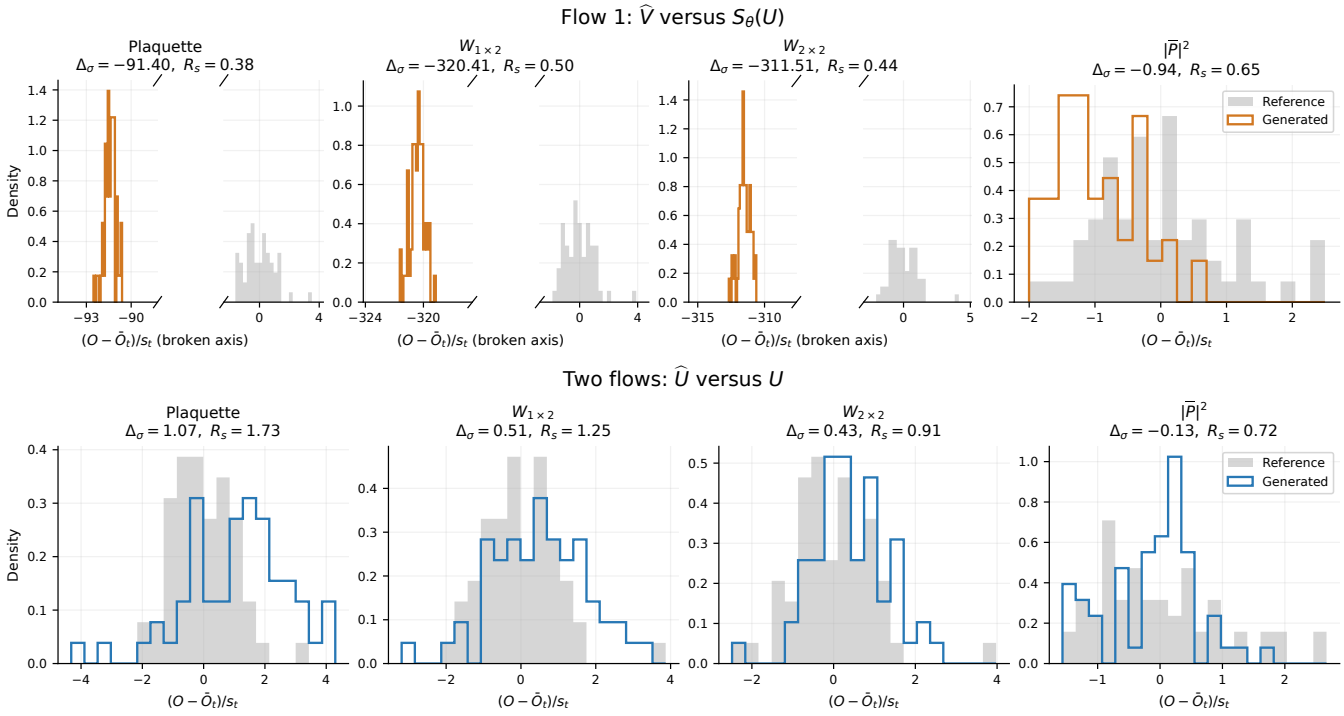


FIG. 4. Stage-by-stage distributions on the same 60 test configurations. Top: Flow 1 output $\widehat{V}$ versus the reference fine configurations after the frozen smoother, $S_{\boldsymbol{\theta}}(U)$, both on 24^4. Bottom: final output $\widehat{U} = F_2(\widehat{V})$ versus original fine references U. Each panel is standardized by its own reference mean and width; annotations give Δ_σ and R_s from Eq. (1.5). The top three loop panels use explicitly marked broken horizontal axes to show the separated distributions; no samples are shifted to create overlap.

[1] J. He, X.-Y. Jin, J. C. Osborn and Y. Zhao, "Neural Field Transformations for Hybrid Monte Carlo: Architectural Design and Scaling," arXiv:2511.02018v1 (2025).

[2] X.-Y. Jin, "Neural Network Gauge Field Transformation for 4D SU(3) gauge fields," PoS LATTICE2023, 040 (2024), arXiv:2405.19692.

[3] C. Morningstar and M. Peardon, "Analytic smearing of SU(3) link variables in lattice QCD," Phys. Rev. D **69**, 054501 (2004), arXiv:hep-lat/0311018.

[4] A. Hasenfratz, E. T. Neil, L. Parato and N. Schwartz, "Renormalization-guided cascade upscaling for lattice field generation," arXiv:2608.28581v1 (2026).